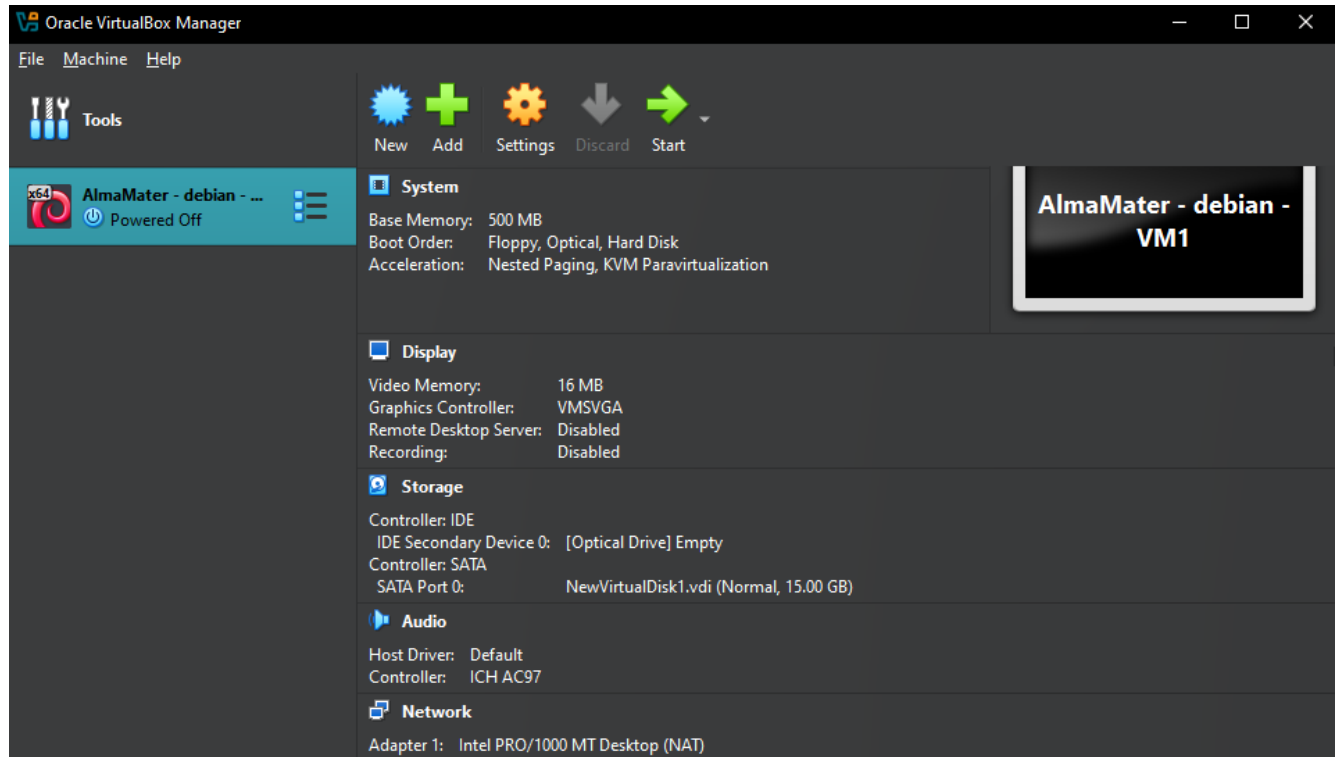


# Domača naloga 2 – Marko Tojagić

## Virtualizacija, omrežja in strežniki

### 1. DELO Z OBSTOJEČIM VIRT. RAČ

Nastavljena količina RAM, omrežni vmesnik: NAT



Prijavljen kot navaden uporabnik, preveril IP naslov VM in preveril če je apache aktiven.

```
AlmaMater - debian - VM1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Debian GNU/Linux 13 markotojagic tty1

markotojagic login: marko
Password:
Linux markotojagic 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
marko@markotojagic:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:fc:d3:cf brd ff:ff:ff:ff:ff:ff
    altname enx080027fcd3cf
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86384sec preferred_lft 75584sec
    inet6 fd17:625c:f037:2:c383:2cef:4242:f751/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86375sec preferred_lft 14375sec
    inet6 fe80::82df:16ca:f192:8498/64 scope link
        valid_lft forever preferred_lft forever
marko@markotojagic:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-03-23 13:34:51 CDT; 50s ago
 Invocation: ebf91e54e63b42048bd8050fdb52072
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 568 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 676 (apache2)
     Tasks: 55 (limit: 492)
   Memory: 7.7M (peak: 7.9M)
      CPU: 142ms
   CGroup: /system.slice/apache2.service
           └─676 /usr/sbin/apache2 -k start
             └─677 /usr/sbin/apache2 -k start
               └─681 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
marko@markotojagic:~$
```

## Zasebni Ipv4 na fizičnem računalniku

```
Administrator: Command Prompt
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::d63a:328e:e929:8076%18
    IPv4 Address. . . . . : 192.168.100.9
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.100.1

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::8b38:e5a6:f4be:7feb%8
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter vEthernet (WSL):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::51de:bd0b:6363:6843%52
    IPv4 Address. . . . . : 172.24.176.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

C:\WINDOWS\system32>
```

VM uporablja IP ki je skrit pred lokalnim omrežjem (NAT).

Dostop preko PuTTY-ja (SSH) in brskalnika prav tako ni mogoč, saj instanca ni v istem (lokalnem) omrežju.

Po ponovnem zagonu sem preveril IP ter se povezal preko PuTTY in brskalnika. Bridged omogoča da je instanca vidna v lokalnem omrežju in je to mogoče.

ReferencesMailingsReviewViewHelpAcrobatPDFelementTell me what you want to do

net can contain viruses. Unless you need to edit, it's safer to stay in Protected View. Enable Editing

marko@markotojagic:~

login as: marko

marko@192.168.100.11's password:

Linux markotojagic 6.12.74+deb13+1-amd64 #1 SMP PREEMPT\_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86\_64

The programs included with the Debian GNU/Linux system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/\*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

marko@markotojagic:~\$

AlmaMater - debian - VMI [Running] - Oracle VM VirtualBox

FileMachineViewInputDevicesHelp

marko@markotojagic:~\$ ip a

1: lo: <LOOPBACK,UP,LOWER\_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet 127.0.0.1/8 scope host lo

valid\_lft forever preferred\_lft forever

inet6 ::1/128 scope host noprefixroute

valid\_lft forever preferred\_lft forever

2: enp0s3: <BRROADCAST,MULTICAST,UP,LOWER\_UP> mtu 1500 qdisc fq\_codel state UP group default qlen 1000

link/ether 08:00:27:fc:d3:cf brd ff:ff:ff:ff:ff:ff

altname enx080027fcd3cf

inet 192.168.100.11/24 brd 192.168.100.255 scope global dynamic noprefixroute enp0s3

valid\_lft 86377sec preferred\_lft 75577sec

inet6 fe80::209:1:dc0a:f32:8436/64 scope link

valid\_lft forever preferred\_lft forever

marko@markotojagic:~\$ \_

5. Virtualni računalnik:

o pravilno izklopite

o spremenite omrežni vmesnik na Bridged (povezan)

6. Po ponovnem zagonu:

o ponovno preverite IP naslov

o ponovno poskusite dostop (SSH in brskalnik)

o primerjajte rezultate z NAT načinom

3. NOV VIRTUALNI STREŽNIK

Marko Tojagic

Not secure 192.168.100.11

Verify it's you

Chat

do

Marko Tojagic



AlmaMater - debian - VMI [Running] - Oracle VM VirtualBox

FileMachineViewInputDevicesHelp

marko@markotojagic:~\$ ip a

1: lo: <LOOPBACK,UP,LOWER\_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet 127.0.0.1/8 scope host lo

valid\_lft forever preferred\_lft forever

inet6 ::1/128 scope host noprefixroute

valid\_lft forever preferred\_lft forever

2: enp0s3: <BRROADCAST,MULTICAST,UP,LOWER\_UP> mtu 1500 qdisc fq\_codel state UP group default qlen 1000

link/ether 08:00:27:fc:d3:cf brd ff:ff:ff:ff:ff:ff

altname enx080027fcd3cf

inet 192.168.100.11/24 brd 192.168.100.255 scope global dynamic noprefixroute enp0s3

valid\_lft 86377sec preferred\_lft 75577sec

inet6 fe80::209:1:dc0a:f32:8436/64 scope link

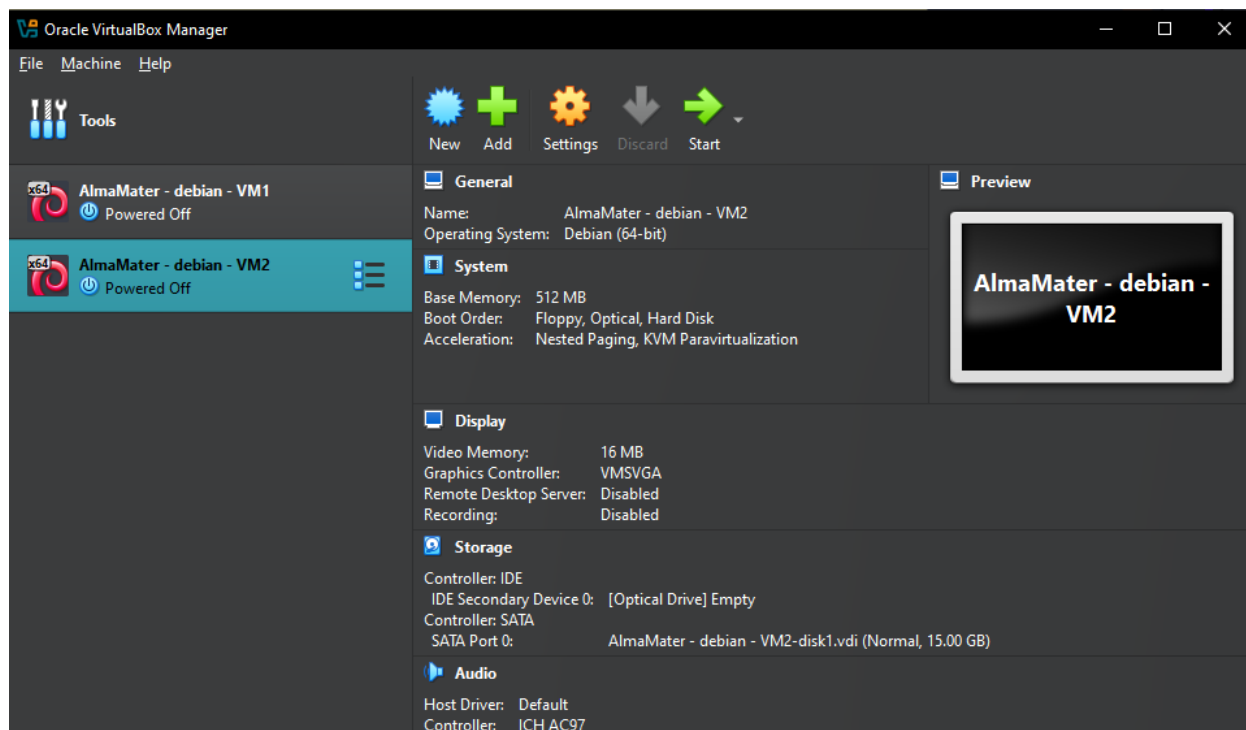
valid\_lft forever preferred\_lft forever

marko@markotojagic:~\$ \_

## 2. NOV VIRTUALNI RAČUNALNIK

Ustvarjen nov virtualni rač.

1 CPU, 512MB RAM, Bridged omrežni vmesnik, ter sem namestil operacijski sistem Debian.



Namestil sem SSH strežnik, Apache spletni strežnik ter preveril delovanje in preveril IP naslov.

```
AlmaMater - debian - VM2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:db:dd:d0 brd ff:ff:ff:ff:ff:ff
    altname enx080027dbdd0
    inet 192.168.100.12/24 brd 192.168.100.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85548sec preferred_lft 74748sec
    inet6 fe80::b707:76c0:50a:2784/64 scope link
        valid_lft forever preferred_lft forever
root@markotojagic:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-03-23 14:00:01 CDT; 5min ago
 Invocation: 48456277cb1a44c6bc327a350c83bcf1
    Docs: https://httpd.apache.org/docs/2.4/
  Process: 1065 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 1068 (apache2)
    Tasks: 55 (limit: 506)
   Memory: 5.3M (peak: 5.5M)
      CPU: 101ms
   CGroup: /system.slice/apache2.service
           └─1068 /usr/sbin/apache2 -k start
             └─1070 /usr/sbin/apache2 -k start
               └─1073 /usr/sbin/apache2 -k start

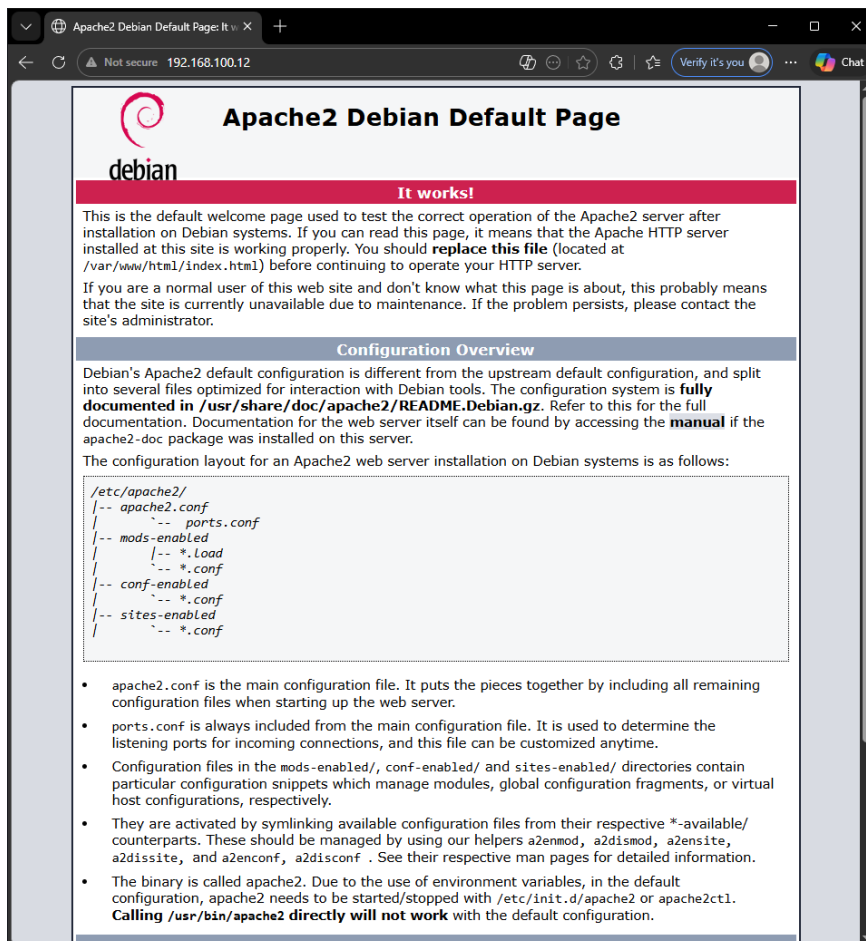
Mar 23 14:00:01 markotojagic systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 23 14:00:01 markotojagic systemd[1]: Started apache2.service - The Apache HTTP Server.
root@markotojagic:~# systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-03-23 13:51:14 CDT; 14min ago
 Invocation: 603c4c6498564a54ab961f689fb0eb51
    Docs: man:sshd(8)
          man:sshd_config(5)
 Main PID: 615 (sshd)
    Tasks: 1 (limit: 506)
   Memory: 2.6M (peak: 3.3M)
      CPU: 69ms
   CGroup: /system.slice/ssh.service
           └─615 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 23 13:51:14 markotojagic systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 23 13:51:14 markotojagic sshd[615]: Server listening on 0.0.0.0 port 22.
Mar 23 13:51:14 markotojagic sshd[615]: Server listening on :: port 22.
Mar 23 13:51:14 markotojagic systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
root@markotojagic:~#
```

Vzpostavil sem SSH povezavo z uporabo aplikacije PuTTY.

```
marko@markotojagic: ~  
login as: marko  
marko@192.168.100.12's password:  
Linux markotojagic 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2  
(2026-03-08) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Mon Mar 23 13:46:34 2026 from 192.168.100.9  
marko@markotojagic:~$
```

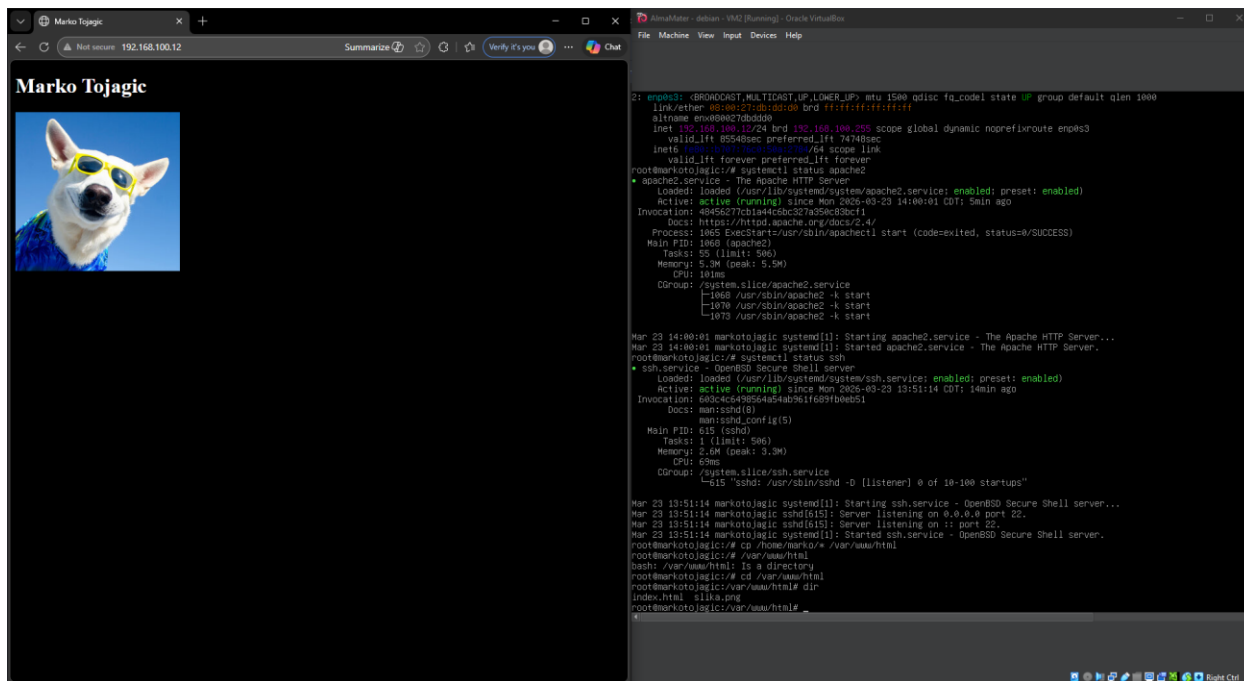
Odpri sem tudi spletno stran strežnika v brskalniku. Prikaže se apache default page.



Uspešno sem poslal iz prvega virt. računalnika datoteki spletne strani v drugi virt. računalnik.

```
root@markotojagic:/home/marko# scp -r /var/www/html/* marko@192.168.100.12:/home/marko/
marko@192.168.100.12's password:
index.html
slika.png
root@markotojagic:/home/marko# _
```

Po prenosu sem kopiral datoteki v /var/www/html/ (v mapo strežnika za spletno stran) in se mi je uspešno pojavila spletna stran v brskalniku.





### 3. VIRUTALNI STREŽNIK V OBLAKU AWS EC2

Ustvaril sem EC2 instanco z OS Debian. Uporabljal sem privzete nastavitve ter sem ustvaril ključ za dostop, saj nisem bil prisoten na predavanju. (Pogledal sem posnetek)

#### Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

##### Name and tags [Info](#)

Name

AlmaMater - debian - VM1

[Add additional tags](#)

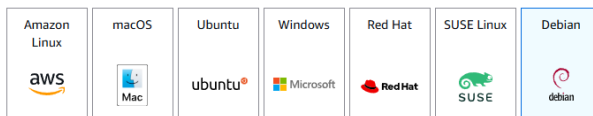
##### ▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

**Quick Start**



[Browse more AMIs](#)  
Including AMIs from AWS, Marketplace and the Community

##### Amazon Machine Image (AMI)

Debian 13 (HVM), SSD Volume Type  
ami-0da1f66573556d917 (64-bit (x86)) / ami-08241d277446b81d7 (64-bit (Arm))  
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

##### Description

Debian 13 (HVM), EBS General Purpose (SSD) Volume Type. Community developed free GNU/Linux distribution. <https://www.debian.org/>

Debian 13 (20260316-2418)

##### Instance type

**t3.micro** Free tier eligible  
Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand SUSE base pricing: 0.012 USD per Hour  
On-Demand RHEL base pricing: 0.0408 USD per Hour On-Demand Linux base pricing: 0.012 USD per Hour  
On-Demand Ubuntu Pro base pricing: 0.0155 USD per Hour On-Demand Windows base pricing: 0.0212 USD per Hour

☒ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

##### ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

almamaterkey

[Create new key pair](#)

##### ▼ Network settings [Info](#)

[Edit](#)

Network [Info](#)

vpc-055c99de93f2a6502

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-3' with the following rules:

☒ Allow SSH traffic from  
Helps you connect to your instance

My IP  
31.15.198.217/32

☐ Allow HTTPS traffic from the internet

To set up an endpoint for remote access, create a web console

Povezal sem se preko SSH (PuTTY) – private key .ppk sem uporabil v Auth sekciji PuTTY nastavitvev, ter sem v Session Hostname vpisal public IPV4 od instance, s tem sem dobil dostop do EC2 instance.

```
admin@ip-172-31-0-60: ~  
login as: admin  
Authenticating with public key "almamaterkey"  
Linux ip-172-31-0-60 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
admin@ip-172-31-0-60:~$
```

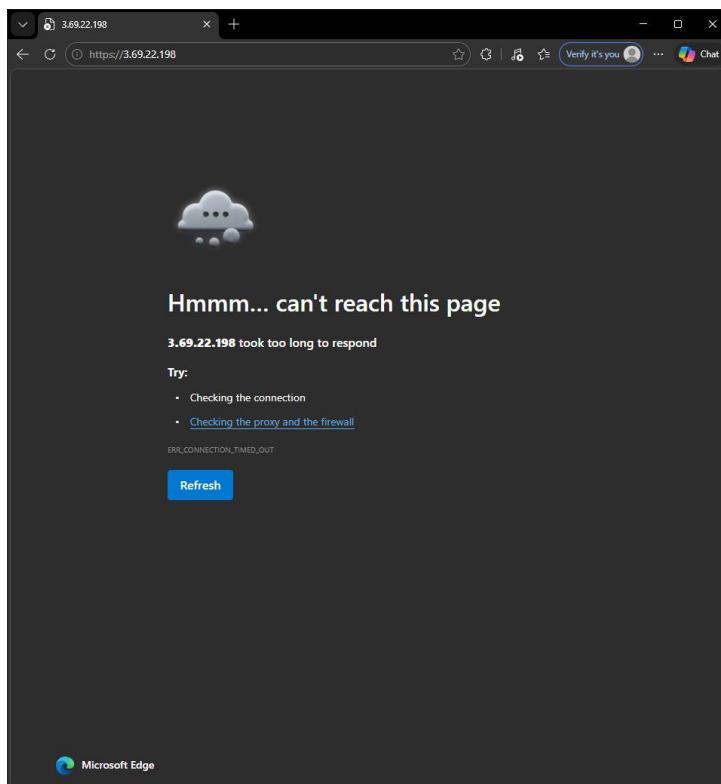
Preveril sem delovanje SSH na strežniku. SSH je bil aktiven.

```
admin@ip-172-31-0-60: ~  
admin@ip-172-31-0-60:~$ systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab  
   Active: active (running) since Mon 2026-03-23 19:36:59 UTC; 6min ago  
  Invocation: 7dbd708f10fc4e0b847de6c3030f31  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Process: 700 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)  
 Main PID: 715 (sshd)  
    Tasks: 1 (limit: 1097)  
  Memory: 7.8M (peak: 9.5M)  
     CPU: 84ms  
   CGroup: /system.slice/ssh.service  
           └─715 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Mar 23 19:36:58 ip-172-31-0-60 systemd[1]: Starting ssh.service - OpenBSD Secur  
Mar 23 19:36:59 ip-172-31-0-60 sshd[715]: Server listening on 0.0.0.0 port 22.  
Mar 23 19:36:59 ip-172-31-0-60 sshd[715]: Server listening on :: port 22.  
Mar 23 19:36:59 ip-172-31-0-60 systemd[1]: Started ssh.service - OpenBSD Secur  
Mar 23 19:40:20 ip-172-31-0-60 sshd-session[789]: Accepted publickey for admin  
Mar 23 19:40:20 ip-172-31-0-60 sshd-session[789]: pam_unix(sshd:session): sessi  
admin@ip-172-31-0-60:~$
```

Po tem sem preveril delovanje apache2 (ni deloval – ni bil inštaliran), ter sem preveril IP naslov.

```
admin@ip-172-31-0-60: ~  
Mar 23 19:36:59 ip-172-31-0-60 sshd[715]: Server listening on :: port 22.  
Mar 23 19:36:59 ip-172-31-0-60 systemd[1]: Started ssh.service - OpenBSD Secure  
Mar 23 19:40:20 ip-172-31-0-60 sshd-session[789]: Accepted publickey for admin  
Mar 23 19:40:20 ip-172-31-0-60 sshd-session[789]: pam_unix(sshd:session): sessi  
admin@ip-172-31-0-60:~$ systemctl status apache2  
Unit apache2.service could not be found.  
admin@ip-172-31-0-60:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default  
    link/ether 0a:71:55:19:0d:f7 brd ff:ff:ff:ff:ff:ff  
    altname enp0s5  
    altname enx0a7155190df7  
    inet 172.31.0.60/20 metric 100 brd 172.31.15.255 scope global dynamic ens5  
        valid_lft 3199sec preferred_lft 3199sec  
    inet6 fe80::871:55ff:fe19:df7/64 scope link proto kernel ll  
        valid_lft forever preferred_lft forever  
admin@ip-172-31-0-60:~$
```

Poskušal sem dostopati do strežnika preko brskalnika ampak ni deloval.

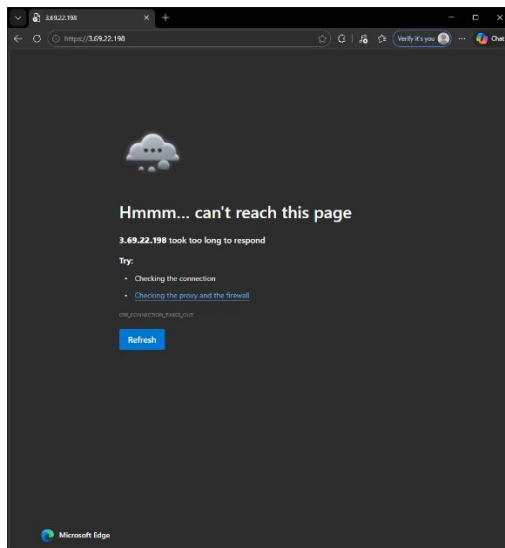


Namestil sem Apache2 strežnik.

```
admin@ip-172-31-0-60: ~
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' ->
'/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean
.service' -> '/usr/lib/systemd/system/apache-htcacheclean.service'.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u2) ...
admin@ip-172-31-0-60:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-03-23 19:51:06 UTC; 22s ago
 Invocation: b7a0922bdc1e34604aa23b21c9eae4098
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1435 (apache2)
     Tasks: 55 (limit: 1097)
    Memory: 5.5M (peak: 5.8M)
       CPU: 42ms
   CGroup: /system.slice/apache2.service
           └─1435 /usr/sbin/apache2 -k start
             └─1436 /usr/sbin/apache2 -k start
               └─1438 /usr/sbin/apache2 -k start

Mar 23 19:51:05 ip-172-31-0-60 systemd[1]: Starting apache2.service - The Apache HTTP Server>
Mar 23 19:51:06 ip-172-31-0-60 apachectl[1434]: AH00558: apache2: Could not reliably determi>
Mar 23 19:51:06 ip-172-31-0-60 systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)
```

Ponovno sem poskusil dostopati preko brskalnika in še vedno ni bilo nič prikazano.



Po tem sem omogočil dostop do spletnega strežnika v Security Group nastavitvah.

sg-0bcf69b3b490bb183 - launch-wizard-1

Actions

#### Details

Security group name  
launch-wizard-1

Security group ID  
sg-0bcf69b3b490bb183

Description  
launch-wizard-1 created 2026-03-23T19:24:39.363Z

VPC ID  
vpc-055c99de93f2a6502

Owner  
161457898876

Inbound rules count  
2 Permission entries

Outbound rules count  
1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

#### Inbound rules (2)

Search



Manage tags

Edit inbound rules

| <input type="checkbox"/> | Name | Security group rule ID | IP version | Type | Protocol | Port range | Source           | Description |
|--------------------------|------|------------------------|------------|------|----------|------------|------------------|-------------|
| <input type="checkbox"/> | -    | sgr-0a13d00ffc491b72d  | IPv4       | HTTP | TCP      | 80         | 0.0.0.0/0        | -           |
| <input type="checkbox"/> | -    | sgr-06a65829332051ff1  | IPv4       | SSH  | TCP      | 22         | 31.15.198.217/32 | -           |

Ponovno sem poskusil dostopati preko brskalnika in se mi je prikazal default page apache2.

Tukaj imam drugi IP za EC2 instanco v brskalniku saj sem vse do tukaj delal 1 dan prej, in se mi je spremenil IP na instanci.

Apache2 Debian Default Page: It works

Not secure 63.181.81.25

Verify it's you

Chat

## Apache2 Debian Default Page

### It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

### Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

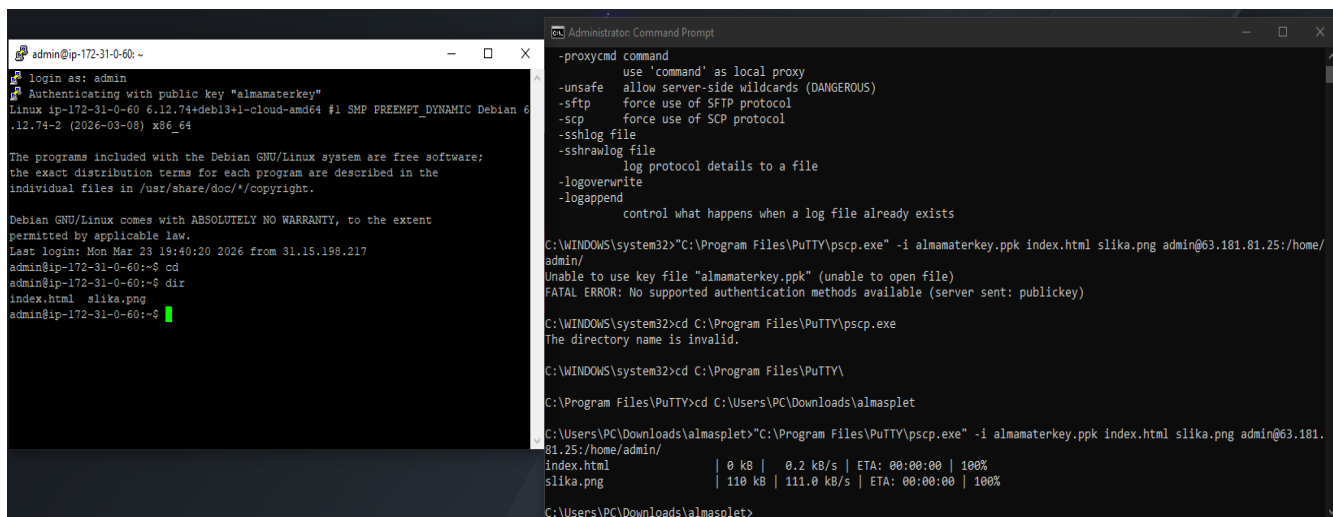
```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf`. See their respective man pages for detailed information.
- The binary is called `apache2`. Due to the use of environment variables, in the default configuration, `apache2` needs to be started/stopped with `/etc/init.d/apache2` or `apache2ctl`. **Calling `/usr/bin/apache2` directly will not work** with the default configuration.

Uporabil sem 2 EC2 instanci za prenos spletnih datotek, to sem storil tako da sem prvo iz fizičnega računalnika prenesel vse na prvo instanco, in kasneje te datoteki iz prve instance skopiral na drugo EC2 instanco.

Tukaj je prikazan prenos iz računalnika v prvo instanco.

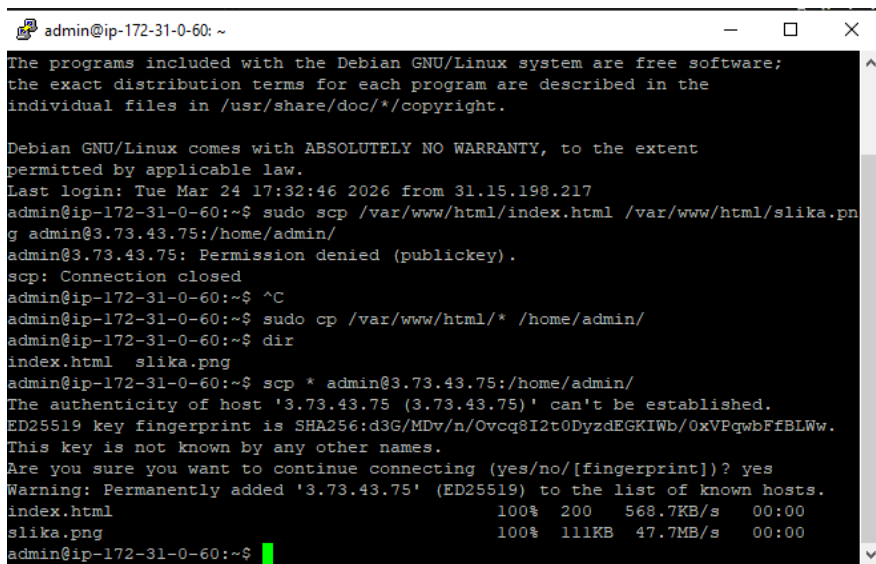
Uporabljal sem pscp (PuTTY scp funkcijo, ker na ta način lahko uporabljam .ppk key za prenos datotek)



```
admin@ip-172-31-0-60: ~  
login as: admin  
Authenticating with public key "almamaterkey"  
Linux ip-172-31-0-60 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Mon Mar 23 19:40:20 2026 from 31.15.198.217  
admin@ip-172-31-0-60:~$ cd  
admin@ip-172-31-0-60:~$ dir  
index.html  slika.png  
admin@ip-172-31-0-60:~$  
  
Administrator: Command Prompt  
-proxycmd command  
-unsafe use 'command' as local proxy  
-sftp allow server-side wildcards (DANGEROUS)  
-sftp force use of SFTP protocol  
-scp force use of SCP protocol  
-sshlog file  
-sshrawlog file  
-logoverwrite log protocol details to a file  
-logappend control what happens when a log file already exists  
  
C:\WINDOWS\system32>"C:\Program Files\PuTTY\pscp.exe" -i almamaterkey.ppk index.html slika.png admin@63.181.81.25:/home/  
admin/  
Unable to use key file "almamaterkey.ppk" (unable to open file)  
FATAL ERROR: No supported authentication methods available (server sent: publickey)  
  
C:\WINDOWS\system32>cd C:\Program Files\PuTTY\pscp.exe  
The directory name is invalid.  
  
C:\WINDOWS\system32>cd C:\Program Files\PuTTY\  
C:\Program Files\PuTTY>cd C:\Users\PC\Downloads\almasplet  
C:\Users\PC\Downloads\almasplet>"C:\Program Files\PuTTY\pscp.exe" -i almamaterkey.ppk index.html slika.png admin@63.181.  
81.25:/home/admin/  
index.html | 0 kB | 0.2 kB/s | ETA: 00:00:00 | 100%  
slika.png | 110 kB | 111.0 kB/s | ETA: 00:00:00 | 100%  
C:\Users\PC\Downloads\almasplet>
```

Tukaj je prikazan uspešni prenos iz prve EC2 instance v drugo. Uporabljal sem Pageant aplikacijo iz PuTTY paketa, kjer sem dodal ključ za drugo instanco, ter v Auth nastavitvah PuTTY-a obkljukal opcijo Agent Forwarding (na ta način sem dosegel da prva EC2 instanca začasno lahko uporablja .ppk private key iz fizičnega računalnika pri prenosu podatkov na drugo EC2 instanco)

Vmes sem mogel tudi v Security Groups (druga instanca) dodati še IP prve instance za SSH povezavo, da bi ta prenos bil uspešen.

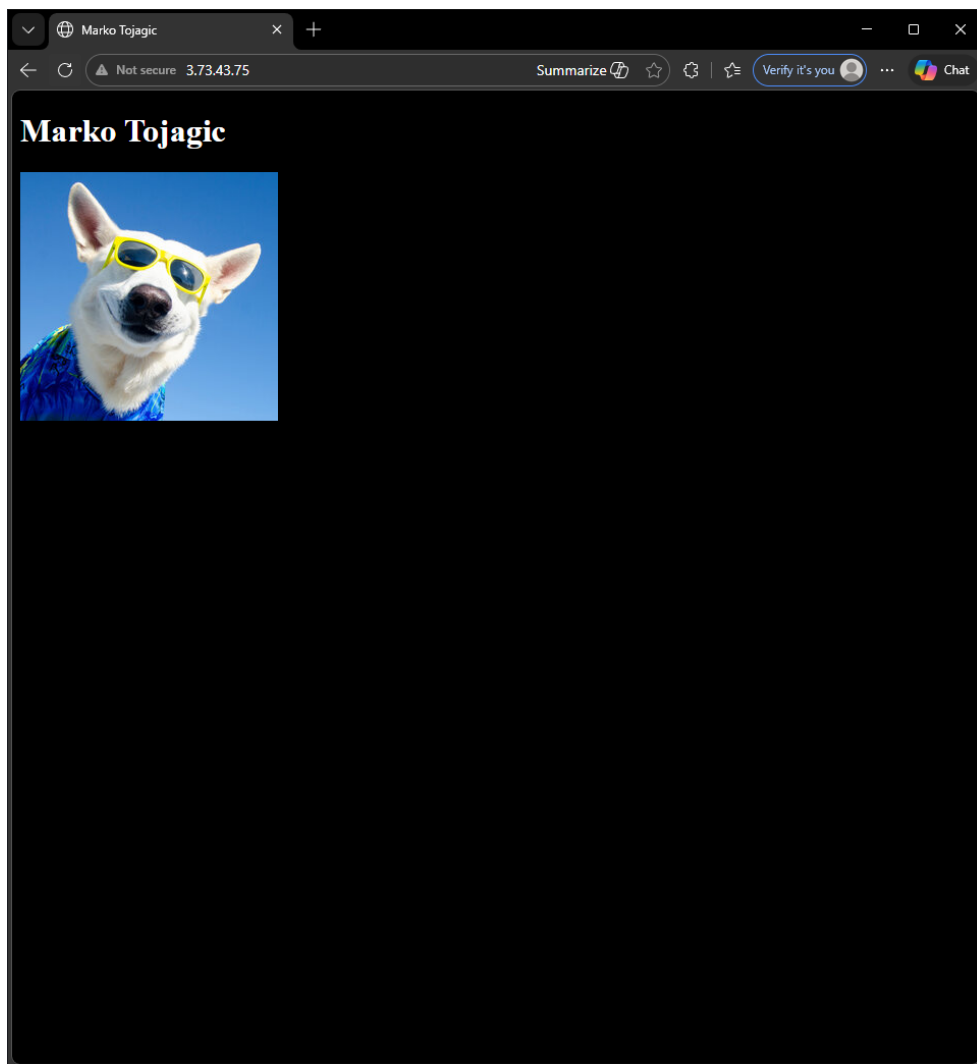


```
admin@ip-172-31-0-60: ~  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Tue Mar 24 17:32:46 2026 from 31.15.198.217  
admin@ip-172-31-0-60:~$ sudo scp /var/www/html/index.html /var/www/html/slika.png  
admin@3.73.43.75:/home/admin/  
admin@3.73.43.75: Permission denied (publickey).  
scp: Connection closed  
admin@ip-172-31-0-60:~$ ^C  
admin@ip-172-31-0-60:~$ sudo cp /var/www/html/* /home/admin/  
admin@ip-172-31-0-60:~$ dir  
index.html  slika.png  
admin@ip-172-31-0-60:~$ scp * admin@3.73.43.75:/home/admin/  
The authenticity of host '3.73.43.75 (3.73.43.75)' can't be established.  
ED25519 key fingerprint is SHA256:d3G/MDv/n/Ovcq8I2t0DyzdEGKIWb/0xVPqwbFfBLWw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '3.73.43.75' (ED25519) to the list of known hosts.  
index.html      100% 200 568.7KB/s 00:00  
slika.png       100% 111KB 47.7MB/s 00:00  
admin@ip-172-31-0-60:~$
```

Sudo scp funkcija ni delovala saj Agent Forwarding posodi .ppk key samo uporabniku in ne root-u.

Po tem sem v drugi instanci EC2 prenesel datoteke iz /home/admin/ v /var/www/html/.

Preveril sem dostop do spletne strani in se mi je uspešno prikazala spletna stran katero sem prenesel.



# ANALIZA

## **Kakšna je razlika med NAT in Bridged omrežjem?**

NAT - VM je skrit v lokalnem omrežju, uporablja pa internet preko host-a (fizičnega računalnika).

Bridged - VM je v istem omrežju kot host, dobi svoj IP in je dosegljiv v lokalnem omrežju.

## **Zakaj v NAT načinu dostop pogosto ne deluje?**

Ker VM ni v lokalnem omrežju, do njega ni mogoče dostopati brez port forwarding-a.

## **Kakšna je razlika med javnim in zasebnim IP naslovom?**

Javni IP je dosegljiv prek interneta.

Zasebni IP se uporablja v lokalnem omrežju, ni neposredno dosegljiv iz interneta.

## **Kakšna je vloga SSH?**

Omogoča varno povezavo s strežnikom in upravljanje strežnika preko šifriranega kanala.

## **Kakšna je vloga Apache strežnika?**

Apache strežnik streže spletne strani in aplikacije prek HTTP/HTTPS protokolov.

## **Kaj omogoča Security Group v AWS?**

Določa pravila za dostop (firewall) do instanc po protokolu, portu in IP naslovu.